EXP-9 Perform File type detection using Autopsy tool

Objective

- Use Autopsy to identify file types in a disk image or logical file set.
- Detect mismatched extensions (file type spoofing) and recover/validate files using header signatures and metadata.
- Demonstrate use of Autopsy ingest modules (File Type Identification, File Carver, Keyword Search) for file-typing and carving.

Prerequisites

- Autopsy installed (version 4.x or later) on Windows or Kali Linux.
- A sample forensic image (.dd, .e01, .img) or a set of files (including some intentionally spoofed filenames, e.g. image.jpg that is actually exe).
- Sleuth Kit installed (Autopsy bundles it usually).
- Administrator privileges to run Autopsy.

Learning Outcomes

After this lab you will be able to:

1. Add a disk image or logical files to Autopsy and run ingest modules for file type detection.
2. Identify files whose extensions do not match their internal signatures (magic bytes).
3. Recover carved files and validate content type and metadata.
4. Generate a report listing suspicious or mismatched files for forensic use.

Tools & Resources

- Autopsy GUI: http://localhost:9999/autopsy (when running)
- Sleuth Kit command-line utilities (optional): fls, icat, tsk_recover
- Sample dataset: Digital Corpora, M57, or lab-provided image with spoofed files.

Procedure

1. Start Autopsy and Create a Case

1. Launch Autopsy: autopsy or open from Start Menu.
2. Create **New Case** → provide case name, examiner, base directory.
3. Click **Add Data Source**.

2. Add Data Source (Image or Logical Files)

- Choose **Disk Image or VM file** for .dd/.img/.E01, or **Logical Files** for a folder of files like test_files.zip.
- Browse and add the file(s).

3. Select Ingest Modules (important ones for file type detection)

- Enable:
    - **File Type Identification (libmagic)**
    - **File Carver** (for carving deleted or unallocated data)
    - **Hash Lookup** (optional — identify known files by hashes)
    - **Keyword Search** (to find suspicious terms)
    - **Extract EXIF, Document Metadata** (optional)
- Start ingest; Autopsy will process and populate results.

4. Inspect File Type Identification Results

1. In left pane, open **Results → File Types** (or view file list).
2. Files are classified by type based on **magic bytes** and extension.
3. Look for entries where **detected type ≠ extension** (e.g., report.pdf detected as ZIP or ELF).

How Autopsy detects file type: uses libmagic (file signature matching) and file headers to determine the probable format regardless of extension.

5. Investigate Mismatched Files

1. Click a suspicious file → **File Metadata / Hex view**.
2. Inspect the first bytes (magic numbers) in hex view:
   - JPEG: FF D8 FF
   - PNG: 89 50 4E 47 0D 0A 1A 0A
   - PDF: %PDF- (25 50 44 46 2D)
   - ZIP / DOCX: 50 4B 03 04
   - EXE (MZ): 4D 5A
3. Confirm the detected file type.

6. Carve Deleted or Fragmented Files

- Use **File Carver** results to recover files that are not in the filesystem metadata.
- Compare carved file signatures to their file names.

7. Validate with Sleuth Kit (optional CLI cross-check)

- List file system entries: fls -r image.dd
- Extract a file by inode: icat image.dd <inode> > extracted.bin
- Run file on extracted file: file extracted.bin to confirm type.

8. Generate Report

1. Use **Generate Report** → choose **HTML/CSV**.
2. Include sections: Mismatched files, carved files, metadata, and notable findings.
3. Save FileTypeReport.html and attach to case folder.

Analysis Checklist (for students)

- Were file extensions and detected types compared?
- Are magic bytes checked for mismatched files?
- Were carved files recovered and validated?
- Is hash lookup used to identify known files (e.g., benign OS files)?
- Is a final report produced with evidence and conclusions?

Example Findings Table (to include in report)

S. No	File Name	Detected Type	Extension	Magic Bytes (hex)	Comment
1	photo.jpg	PNG image	.jpg	89 50 4E 47 0D 0A 1A 0A	Spoofed extension
2	invoice.pdf	ZIP archive	.pdf	50 4B 03 04	Possibly DOCX or ZIP payload

Viva Questions

1. How does Autopsy determine a file's type?
2. What are magic numbers? Give examples.
3. How do you handle files with spoofed extensions in an investigation?
4. What is file carving and when is it needed?
5. How can hash lookup help during file-type analysis?
6. Why might a valid PDF begin with 50 4B 03 04?
7. How would you prove in court that a file was intentionally disguised?